

Simple Harmonic Motion Calculations: A Beginner's Guide

Lesson 6: Solving Problems: Spring-Mass Systems

Solving Problems: Spring-Mass Systems - Study Notes

Key Concepts

- **Simple Harmonic Motion (SHM):** The oscillatory motion of a mass attached to a spring, where the restoring force is proportional to the displacement from equilibrium.
- **Period (T):** The time it takes for one complete oscillation. Measured in seconds (s).
- **Frequency (f):** The number of oscillations per unit time. Measured in Hertz (Hz), where 1 Hz = 1 oscillation/second. Frequency is the inverse of the period ($f = 1/T$).
- **Spring Constant (k):** A measure of the stiffness of the spring. Higher k means a stiffer spring. Measured in Newtons per meter (N/m).
- **Amplitude (A):** The maximum displacement of the mass from its equilibrium position. Measured in meters (m).

Summary

Spring-mass systems provide a great example of Simple Harmonic Motion. When a mass is attached to a spring and displaced from its equilibrium position, the spring exerts a restoring force that pulls the mass back. This force causes the mass to oscillate back and forth around the equilibrium point. The period and frequency of this oscillation depend primarily on the mass and the spring constant – a heavier mass or a stiffer spring will result in a shorter period (and higher frequency). Understanding these relationships allows us to predict and calculate key parameters like the period, frequency, and maximum speed of the oscillating mass.

The energy in a spring-mass system is constantly being exchanged between potential energy (stored in the spring when stretched or compressed) and kinetic energy (energy of motion of the mass). At the maximum displacement (amplitude), all the energy is potential. As the mass passes through the equilibrium position, all the energy is kinetic. The total mechanical energy of the system remains constant (assuming no energy loss due to friction or damping).

Important Points

- **Formula for Period (T):** $T = 2\pi\sqrt{m/k}$
- * T = Period (seconds)
- * m = Mass (kilograms)
- * k = Spring Constant (N/m)
- **Formula for Frequency (f):** $f = 1/T$ or $f = \frac{1}{2\pi}\sqrt{k/m}$
- **Maximum Speed (v_{max}):** $v_{\text{max}} = A\omega$
- * A = Amplitude (meters)
- * ω = Angular Frequency = $\sqrt{k/m}$ (radians/second)
- **Angular Frequency (ω):** $\omega = \sqrt{k/m}$
- **Units are crucial:** Ensure you use consistent SI units (kg, m, s, N).

- **Equilibrium Position:** The point where the spring is neither stretched nor compressed, and the net force on the mass is zero.
- **Assumptions:** These calculations assume an ideal spring (obeys Hooke's Law) and no damping forces (friction, air resistance).

Examples

Example 1: Calculating Period

A 2 kg mass is attached to a spring with a spring constant of 400 N/m. The mass is pulled 0.1 m from equilibrium and released. What is the period of oscillation?

- **Given:** $m = 2 \text{ kg}$, $k = 400 \text{ N/m}$
- **Formula:** $T = 2\pi\sqrt{m/k}$
- **Calculation:** $T = 2\pi\sqrt{2/400} \approx 0.005 \text{ s}$
- **Answer:** The period of oscillation is approximately 0.444 seconds.

Example 2: Calculating Maximum Speed

Using the same system as above ($m = 2 \text{ kg}$, $k = 400 \text{ N/m}$), the mass is pulled 0.1 m from equilibrium and released. What is the maximum speed of the mass?

- **Given:** $m = 2 \text{ kg}$, $k = 400 \text{ N/m}$, $A = 0.1 \text{ m}$
- **Step 1:** Calculate Angular Frequency ($\omega = \sqrt{k/m} = \sqrt{400 \text{ N/m} / 2 \text{ kg}} = 200 \text{ rad/s}$)
- **Step 2:** Calculate Maximum Speed ($v_{\text{max}} = A\omega = 0.1 \text{ m} \cdot 200 \text{ rad/s} = 14.14 \text{ m/s}$)
- **Answer:** The maximum speed of the mass is approximately 1.414 m/s.

Quick Review

1. What is the relationship between period (T) and frequency (f)?
2. How does increasing the mass of an object in a spring-mass system affect the period of oscillation?
3. What is the spring constant, and what does a higher value indicate?
4. Calculate the frequency of a spring-mass system with a mass of 0.5 kg and a spring constant of 50 N/m.
5. A mass oscillates with a period of 2 seconds. What is its frequency?

Further Reading

- **Hooke's Law:** Explore the mathematical relationship between force and displacement in a spring.
- **Damped Harmonic Motion:** Investigate what happens when damping forces (like friction) are present in a spring-mass system.
- **Driven Harmonic Motion:** Learn about systems where an external force drives the oscillation.
- **Energy Conservation:** Delve deeper into the exchange of potential and kinetic energy in the system.
- **Simple Pendulums:** Compare and contrast the motion of a spring-mass system with a simple pendulum.

